Basal Ganglia: Cram Sheet

NSC 4351 · Motor: Basal Ganglia lecture · red = he flagged it / trap

- **BG = a brake.** Too much brake → **hypo**kinetic (PD: can't start). Too little → **hyper**kinetic (HD, chorea, ballismus, dystonia, tics: can't stop).
- **BG lesions NEVER cause weakness.** No output to spinal cord; acts only via thalamus → cortex. (Mother-in-law in the passenger seat: can only nag.)
- **Moves while trying to hold STILL = BG.** Cerebellum only misbehaves DURING movement.
- **Same strategy as cerebellum:** all inputs excitatory, one output (GPi/SNr) inhibitory.
- **GUARANTEED Q: find the caudate** (hugs the lateral ventricle).
- **Hemiballismus = CONTRALATERAL subthalamic nucleus.**
- **Decerebrate → decorticate = IMPROVING** (midbrain recovered).
- Direct/indirect wiring **not tested**: just direct = UP, indirect = DOWN.

1 · Parts & Picture ID

Structure	Find it	Key fact
Caudate	C-shape on lateral ventricle: head (front), body (top), tail (temporal horn)	Input; cognitive / eye / emotion loop
Putamen	Lateral to internal capsule, under insula	Input; motor loop
Globus pallidus	Medial to putamen; GPe (outer) + GPi (inner)	**GPi = main output**
Subthalamic n. (STN)	Tiny, under thalamus	Hemiballismus
Substantia nigra	Midbrain, dark stripe	SNc = dopamine; SNr = output
N. accumbens	Under caudate head	Limbic / reward
Internal capsule	White band between caudate & putamen	Cortex ↔ body axons
Basal forebrain (circled)	Under striatum, below ant. commissure	ACh; NOT motor BG; easy to confuse

Names: Striatum = caudate + putamen (striped by cell bridges). Lentiform = putamen + GP. **Neostriatum** = caudate + putamen; **paleostriatum** = GP (quiz Q3).
Trap: posterior coronal cut shows caudate TWICE: body (top) + tail (dot by temporal horn, above hippocampus). Amygdala sits at tail's end.

2 · Midbrain & Dopamine

Label	What
SNc (compacta)	Packed dopamine cells (neuromelanin, black); **die in PD**
SNr (reticulata)	Mesh of fibers (reticule = mesh), no cell bodies; output like GPi
VTA	Midline; dopamine for reward / emotion
CP / RN / ML	Cerebral peduncle (ventral) / red nucleus (round, behind SN) / medial lemniscus

Pathway	Route	Job
Mesostriatal	SNc → striatum	Movement
Mesocortical	VTA → PFC/cortex	Cognition, attention
Mesolimbic	VTA → accumbens	Reward, addiction

3 · The Circuit

Cortex (+glu) → striatum → GPi/SNr (−GABA) → thalamus (+) → cortex
More GPi output = LESS movement. Less GPi output = MORE movement.

- **Order of flow = anatomy outside in:** cortex → caudate/putamen → GPe → GPi → thalamus.
- **Medium spiny neurons** = main striatal cell (GABA). **Die in HD.** SNc dopamine cells die in PD.

	Putamen loop	**Caudate loop**
Input	Motor + somatosensory cx	Association, FEF, limbic cx
Thalamus	**VL** → motor cx	**VA** → prefrontal cx
Job	Movement (evolved 1st)	Cognitive / emotional side (2nd)
Disease	Motor signs PD/HD	Cognitive + emotional deficits PD/HD

	Direct (D1)	**Indirect (D2)**
Cell	GABA + dynorphin	GABA + enkephalin
Route	Str → GPi → thal	Str → GPe → STN → GPi → thal
Effect	**↑ movement**	**↓ movement**
Dopamine	Excites D1	Inhibits D2

Dopamine pushes movement UP through both doors. Lose it (PD) → both swing to less movement.

4 · BG Jobs & Timeline

- **Posture first:** BG predict the shift and set posture BEFORE the move (baby bending for a toy).
- **Runs well-learned programs** (walking), not novel tasks. Needs **proprioceptive** input.
- **Off in sleep** (nothing to control), like cerebellum.

Step	Who
1 Plan	DLPFC/association plans; **BG initiates posture**; lateral cbllm refines; thalamus coordinates; premotor/motor executes
2 Execute	Pyramidal + extrapyramidal tracts → α-MN (final common path)
3 Monitor	Intermediate cerebellum
4 Feedback	Back to motor cx and DLPFC (final judgment)

5 · Disorders

	Parkinson's (too much brake)	**Huntington's (too little)**
Lost cell	SNc dopamine	Medium spiny neurons
Failure	Striatum fails to inhibit GPi → **GPi overactive** (Q6)	GP fails to inhibit thalamus (Q7)
Signs	Brady-hypokinesia, rest tremor, rigidity, **lost postural responses**, bradyphrenia	Chorea, then depression, cognitive decline; family hx

- **Never pat a PD patient's back:** can't load the push-back program → falls. "Can't get out of the chair" = lost postural program, not weakness.
- **Freezing:** "step over my foot" works (non-automatic route).
- **DBS:** STN or GPi equally effective (advanced PD). Also stops tremor.
- **Chorea** = rapid, irregular, dance-like. **Ballismus** = big flinging. **Tic** = shock-like face/neck/voice. **Tremor** = rhythmic oscillation.
- **Dystonia** = sustained patterned contraction, twisting, fixed posture (DYT1 man: twisted, buff, can't walk).
- **Cervical dystonia:** head pulled; **gone in sleep**; **sensory trick** (fingertip on cheek) fixes it; **Botox** ~6 wk. All work by changing sensory input.
- **Hemiballismus:** lupus vasculitis → stroke in **contralateral STN**.
- **Sydenham's:** movement = **Sydenham's chorea**; disease = **rheumatic fever** (post-strep); death = **rheumatic heart disease**. Answer the layer asked.
- **Stereotypy** (hand flapping when excited, stops if distracted) = NOT BG disease → "something else" (Q8).
- **Functional gait** (runs + walks backward fine, can't walk forward) = "something else" (Q1). Fits no anatomy.
- **Tics** are common; touchdown dances are weird, not disease.

6 · Brainstem Tracts & Posturing

Tract	From	Does
Rubrospinal	Red nucleus (midbrain)	**Arm flexors** (red = bend)
Vestibulospinal	Vestibular nuclei	**Leg extensors**; arms out when falling
Tectospinal	Superior colliculus	**Neck**: turn head to stimulus
Reticulospinal	Pons/medulla RF	Backup **extension** of all limbs

	Lost	Arms	Legs
Decorticate	Cortex only; midbrain + RN OK	**Flexed** (rubro)	Extended
Decerebrate (worse)	Cortex + midbrain + RN	**Extended** (reticulo)	Extended

EMT Q: extended → arms bend an hour later = midbrain recovering. Work hard on this patient. Cat falling: head finds ground (tecto) → limbs extend (vestibulo).

7 · Vignette Decoder

Clue	Answer
Not weak, can't hold still	Basal ganglia (not cerebellum)
Can't start, falls when pushed, rest tremor	Parkinson's (GPi overactive)
Dance-like moves + depression + family hx	Huntington's
Flinging one side, lupus	Contralateral STN stroke
Strep → arm movements → heart death	Sydenham's / rheumatic fever
Posture gone in sleep, light touch fixes	Dystonia (cervical)
Arms flexed, legs extended	Decorticate (midbrain intact)
All four extended	Decerebrate (worse)
Main BG output? (Q4)	GPi

8 · In-Class Quiz (Oct 22)

Q	Answer
1 Gait video (can't walk forward)	Something else: functional
2 Strep, chorea, heart death at 23	Sydenham's chorea / rheumatic fever
3 Box under paleostriatum	Globus pallidus
4 Main BG output	GPi
6 PD: what fails?	Caudate/putamen fails to inhibit GPi
7 HD: what fails?	GP fails to inhibit thalamus
8 Girl flaps hands seeing dad	Something else: stereotypy

Pattern: "something else" was right repeatedly. Check the movement truly fits the named disorder's definition before picking it.